NORTHERN UNIVERSITY OF BUSINESS AND TECHNOLOGY, KHULNA

User Manual

AI Smart Campus System

Installation and Operational Guide

Course: CSE 4204 - Mobile Computing Lab

Week 12 Final Submission

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Submission date: 22 October 2026

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1. System Overview

AI Smart Campus System is a browser-based campus platform with student, faculty and administrator workspaces after authentication and approval. Navigation and permissions depend on the signed-in role.

2. Installation and Setup

Step	Action
1	Install PHP 8.2+, Composer, MySQL, Node.js and npm.
2	In backend, copy .env.example to .env; configure database values and protected AI keys; run composer install.
3	Run php artisan key:generate, php artisan migrate, php artisan storage:link and php artisan db:seed as appropriate.
4	Run php artisan serve to start the API.
5	In frontend, run npm install and npm run dev. Use npm.cmd run dev if PowerShell blocks npm.ps1.
6	Open the Vite address printed by the development server; reviewed local evidence used localhost:5173.

3. Login and User Roles

Open Login, enter a registered email and password, then select Sign In. Public registration accepts Student or Faculty roles; administrator self-registration is rejected. New accounts may require administrator approval. Use Forgot Password to request reset instructions.

Role	Main capabilities
Student	Dashboard, profile, academic data/exports, tasks, resources, messages, services, recommendations and AI Assistant.
Faculty	Dashboard, assigned course workspace, record entry, scoped monitoring, notices, services and AI Assistant.
Administrator	User approval, departments, courses, notices, campus operations and institution-wide monitoring.

4. Student Workflow

1. Sign in with an approved student account. 2. Open Dashboard to view personal academic information. 3. Use Campus Services for routines, exams, events, fees, tickets, leave/reschedule and library actions where permitted. 4. Open Course Recommendations and create a study-plan task where offered. 5. Open Messages to view/read notices or download authorized attachments. 6. Use My Profile to update permitted details, password or photo. 7. Download transcript or attendance exports. 8. Open AI Assistant, enter a question and submit.

5. Faculty Workflow

1. Sign in with an approved faculty account. 2. Open Academic Management and select an assigned course workspace. 3. Manage authorized enrollment, attendance, assessments, grades and performance. 4. Open Student Monitoring to inspect assigned-course students. 5. Run analysis and review score, reasons, prediction, advice and source. Results can be labelled BASELINE if the external provider is unavailable. 6. Manage authorized notices/services and use the assistant.

6. Administrator Workflow

1. Sign in as an administrator. 2. Use Manage Users for pending accounts, approval status and all-user review. 3. Manage departments. 4. Use Academic Management for course and institution-wide academic records. 5. Publish, target, archive or remove notices. 6. Manage authorized schedules, exams, events, fees and library/service records. 7. Review Student Monitoring and Risk Alerts.

7. Feature Guide

Feature	How to use
Tasks	Create, view, update or delete personal tasks according to authorization.
Learning resources	Browse eligible resources; authorized staff may upload/manage them.
Notices	Messages shows applicable notices; read/download/archive/manage actions are role-scoped.
Academic management	Select a course and use workspace controls for enrollment and records.
Campus operations	View data or submit/manage operations according to role.
AI Assistant	Student/faculty only: type a question and wait for a response.
Risk analysis	Faculty/admin: select a permitted student and inspect persisted result.
Exports	Download authorized transcript or attendance records.

8. Screenshots

No current, submission-ready screenshot files were available in screenshots/ at documentation time; that folder contains only an index requesting future captures. This manual therefore contains no invented or stale screenshots. Capture sanitized desktop and mobile screens before submission and never show passwords, tokens, API keys or private student data.

9. Troubleshooting

Symptom	Action
Cannot log in	Check credentials, approval, backend availability and auth rate-limit cooldown.
Messages database error	Run pending Laravel migrations, then refresh.
AI unavailable	Check key, model, network and quota. Keys belong only in backend/.env.

Symptom	Action
No page access	Confirm role and faculty course assignment; access is intentionally scoped.
Upload fails	Check storage link, permissions and file validation.
Frontend cannot reach API	Check Vite API URL, Laravel server and CORS_ALLOWED_ORIGINS.
Reset email absent	Verify mail configuration and backend logs.

10. Deployment Notes

Point the web root to backend/public, serve frontend/dist with SPA fallback, enable HTTPS, restrict CORS origins, use APP_DEBUG=false, run production migrations with --force, configure queue workers when notice delivery is enabled, and keep .env, logs, backups and source maps out of the public web root.